Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,749 earthquake events in Myanmar from 1970-03-10 to 2026-08-02. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.037 with magnitude of completeness Mc=4.7.

Total Events: 9,749

Date Range: 1970-03-10 to 2026-08-02

Max Magnitude: M7.7

Mean Depth: 25.7 km

b-value: 1.037

a-value: 7.690

Mc (completeness): 4.7

2. Earthquake Map

3. Seismic Clustering Analysis

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4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 44 dams

High Risk: 116 dams

Moderate Risk: 45 dams

Low Risk: 49 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist Fault	Risk	Grade
Dam near 21.74N 95.85E	nan	0.2718	7.4 km	10.0	Critical
Dam near 22.09N 95.86E	nan	0.2995	5.7 km	10.0	Critical
Dam near 21.77N 95.89E	nan	0.3257	2.7 km	10.0	Critical
Dam near 21.47N 95.87E	nan	0.2815	6.0 km	9.7	Critical
Dam near 20.85N 96.03E	nan	0.4813	8.3 km	8.8	Critical
Meikhtila	nan	0.2412	8.2 km	8.8	Critical
Dam near 21.34N 95.81E	nan	0.2263	12.2 km	8.8	Critical
Natkar	nan	0.2443	7.9 km	8.7	Critical
Dam near 21.37N 95.80E	nan	0.2207	13.4 km	8.7	Critical
Dam near 20.72N 95.90E	nan	0.2665	5.6 km	8.7	Critical
Dam near 21.04N 95.83E	nan	0.2270	10.9 km	8.7	Critical
Dam near 21.05N 95.83E	nan	0.2265	11.0 km	8.7	Critical
Dam near 20.47N 96.08E	nan	0.5000	9.9 km	8.6	Critical
Dam near 20.60N 95.90E	nan	0.2464	7.4 km	8.6	Critical
Taungnyo	Taungnyo	0.3054	1.2 km	8.6	Critical
Nat Thar Taw	Nat Thar Taw	0.1903	20.2 km	8.5	Critical
Myotha	Ku Htaw	0.1849	20.6 km	8.5	Critical
Thaphan Chaung	Thaphan Chaung	0.2197	11.5 km	8.2	Critical
Zeedaw	Nyaung Oat	0.2075	15.3 km	8.2	Critical
Dam near 17.55N 96.37E	nan	0.1022	9.6 km	8.2	Critical
Male-Nattaung	Male-Nattaung	0.2360	18.6 km	8.1	Critical
Dam near 20.73N 95.83E	nan	0.2110	13.6 km	7.9	Critical
Nyaunggone	nan	0.2104	13.7 km	7.9	Critical
Paunggadaw	Paunggadaw	0.1488	33.9 km	7.9	Critical
Kun Chaung	Kun Chaung	0.4644	0.9 km	7.7	Critical
Kintat	Mu	0.1669	21.0 km	7.7	Critical
North Pinle	Pinle	0.1427	36.1 km	7.7	Critical
Phyu Chaung	Pyu Chaung	0.4707	1.3 km	7.6	Critical
South Pinle	Pinle	0.1383	38.1 km	7.6	Critical
Kabaung	Kabaung Chaung	0.3335	9.2 km	7.6	Critical

5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 34.92

c (time offset): 16.408 hours

p (decay exponent): 0.772

Expected M>=3 (7d): 4

Expected M>=3 (30d): 18

Expected M>=3 (90d): 51

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit